

云计算实验三报告

作业 1:

```
[root@10-23-70-56 ~]# hdfs dfsadmin -report
Configured Capacity: 633648111616 (590.13 GB)
Present Capacity: 601278963712 (559.98 GB)
DFS Remaining: 601278877696 (559.98 GB)
DFS Used: 86016 (84 KB)
DFS Used%: 0.00%
Under replicated blocks: 0
Blocks with corrupt replicas: 0
Missing blocks: 0
Missing blocks (with replication factor 1): 0
Pending deletion blocks: 0

-----
report: Access denied for user root. Superuser privilege is required
[root@10-23-70-56 ~]#
```

作业 2:

```
[root@10-23-70-56 ~]# touch info.txt
[root@10-23-70-56 ~]# echo "云计算实验课:10211900416-郭夏辉" > info.txt
[root@10-23-70-56 ~]# cat info.txt
云计算实验课:10211900416-郭夏辉
[root@10-23-70-56 ~]# hadoop fs -mkdir /test
[root@10-23-70-56 ~]# hadoop fs -put info.txt /test
[root@10-23-70-56 ~]# hadoop fs -ls /test
Found 1 items
-rw-r--r--  3 root supergroup      41 2023-10-11 16:32 /test/info.txt
[root@10-23-70-56 ~]# hadoop fs -cat /test/info.txt
云计算实验课:10211900416-郭夏辉
```

```
[root@10-23-70-56 ~]# hadoop fs -rm -f /test/info.txt
23/10/11 16:36:57 INFO fs.TrashPolicyDefault: Moved: 'hdfs://Ucluster/user/root/.Trash/Current/test/info.txt' to trash at: hdfs://Ucluster/user/root/.Trash/Current/test/info.txt
[root@10-23-70-56 ~]# hadoop fs -rmdir /test
```

作业 3:

```
[root@10-23-70-56 ~]# hadoop fs -mkdir /input
[root@10-23-70-56 ~]# hadoop fs -put /root/hadoop-2.6.0/conf/* /input
[root@10-23-70-56 ~]# hadoop jar /root/hadoop-2.6.0/share/hadoop/mapreduce/hadoop-mapreduce-examples-2.8.5.jar wordcount /input /output
```

由于结果过长，我把它存到了 test.txt:

```
[root@10-23-70-56 ~]# hadoop fs -cat /output/part-r-00000 > test.txt
[root@10-23-70-56 ~]# head test.txt
!=      3
""      7
""      4
"$HADOOP_CLASSPATH"      1
"$HADOOP_HEAPSIZE"      1
"$JAVA_HOME"      2
"$YARN_HEAPSIZE"      1
"$YARN_LOGFILE"      1
"$YARN_LOG_DIR"      1
"$YARN_POLICYFILE"      1
[root@10-23-70-56 ~]#
```

作业 4:

这个实验我是在云主机里进行的，因为本地的 windows 之 type 命令实在是太难用了，不如 linux 的 cat 那样简洁好用。

map.py:

```
import sys
import time
Start = time.time()

for linei in sys.stdin:
    linei = linei.strip()
    words = linei.split()
    for word in words:
        print("%s\t%s" % (word, 1)) # map

End = time.time()
print('map costs', (End - Start), 's')
```

reduce.py:

```
import sys
import time
import re
Start = time.time()
mr_dic = {}

for linei in sys.stdin:
    linei = linei.strip()
    parts = re.split(r'\t| ', linei)

    if len(parts) != 2:
        continue

    word, counting = parts

    try:
        counting = int(counting)
    except ValueError:
        continue

    if word in mr_dic:
        mr_dic[word] += 1
    else:
```

```
mr_dic.setdefault(word, 1)

for key, value in mr_dic.items():
    print('%s\t%s' % (key, value))

End = time.time()
print('reduce costs', (End - Start), 's')
```

```
[root@10-23-70-56 ~]# ls
hadoop  hadoop-2.6.0  hive  info.txt  java  map.py  reduce.py  test1
[root@10-23-70-56 ~]# cat /hadoop/*
cat: '/hadoop/*': No such file or directory
[root@10-23-70-56 ~]# clear
[root@10-23-70-56 ~]# cat hadoop/* > test1
[root@10-23-70-56 ~]# cat test1 | python map.py > test2
[root@10-23-70-56 ~]# tail -1 test2
('map costs', 0.0058629512786865234, 's')
[root@10-23-70-56 ~]# cat test2 | python reduce.py > test3
[root@10-23-70-56 ~]# tail -1 test3
('reduce costs', 0.020557165145874023, 's')
```

map 用时 0.0586s reduce 用时 0.020557s,还是比较快的。

```
[root@10-23-70-56 ~]# grep -w "!=" test3
!=      3
```

试了几次之后，发现结果与之前的那个 test.txt 中是一样的。